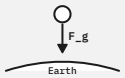
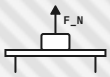
Say it, then write it: who pushes?

Language goal: I can **name** who exerts a force.



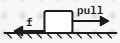
gravity
gravedad · gravidade
Earth pulls down



normal force
fuerza normal · força normal
a surface pushes out



tension
tensión · tração
a rope pulls



friction
fricción · atrito
rubbing that resists sliding

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: Earth · table · rope · floor · newtons

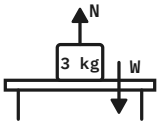
Gravity on the box comes from ____. The normal force comes from the ____. Tension comes from the ____. Friction comes from the ____. We measure every force in ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The ____ exerts a ____ force on the box.
The force points ____ because ____ .

3 · SOLVE WITH THE STEPS



A 3 kg box sits on a table. Use g ≈ 10 N/kg.

STEP 1 · WEIGHT
F_g = ____ × ____ = ____ N

STEP 2 · SUPPORT
normal force = ____ N, pointing ____

STEP 3 · SAY IT
"Earth pulls ____ N down; the table pushes ____ N up."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ Every force has a source. The source of ____ is ____ .
☐ "The force of motion" is not a force because ____ .
☐ First I name the ____, then I draw the ____ .

I said it ☐ · I wrote it ☐

K2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Mass and weight

10 H

DIRECTIONS · Fill all four boxes for each term.

Mass

DEFINITION · MY WORDS

UNITS + TOOL

unit ____ · measured with ____

EXAMPLES

DOES IT CHANGE ON THE MOON?

Weight

DEFINITION · MY WORDS

FORMULA

F_g = ____ × ____

EXAMPLE CALCULATION

MASS vs. WEIGHT MIX-UP

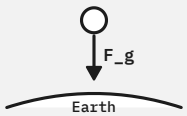
One sentence using both terms: _____

K2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: five forces

Language goal: I can **name** who exerts a force.



↓ Earth

gravity F_g

gravedad · gravidade

Earth pulls down

my language: _____

use it: _____



↑ surface

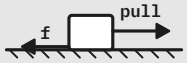
normal F_N

fuerza normal · força normal

a surface pushes ⊥

my language: _____

use it: _____



← rub

friction f

fricción · atrito

along the surface, opposes slip

my language: _____

use it: _____



rope →

tension T

tensión · tração

a rope or string pulls

my language: _____

use it: _____



hand →

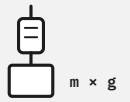
applied

fuerza aplicada · força aplicada

a push or pull you give

my language: _____

use it: _____



m × g

weight

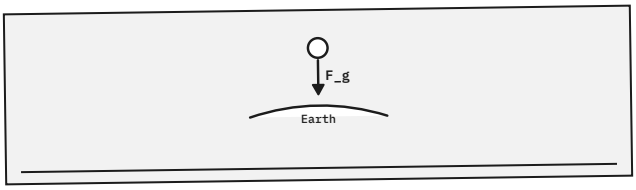
peso · peso

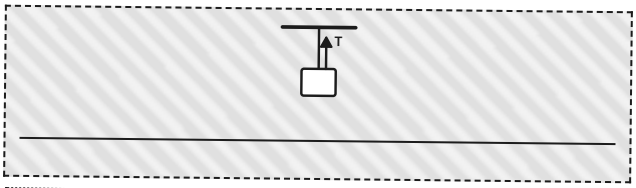
gravity force in newtons

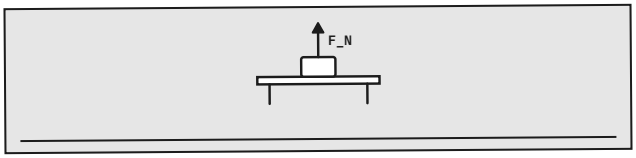
my language: _____

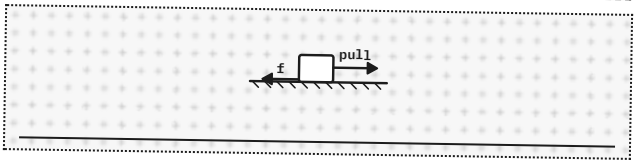
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner □

K2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

10 T

Who pushed the crate?

A crate slid across the stockroom floor. Name every force — and who exerted it.

EXHIBIT A · THE SCENE

A 20 kg crate is pulled right by a rope with 60 N. The floor is rough. It slides right. Use g ≈ 10 N/kg.

Source: rope → ____

Source: floor (up) → ____

Source: floor (along) → ____

Source: Earth → ____

ANALYSIS 1 · WEIGHT

$F_g = m \times g =$ ____ N, pointing ____

ANALYSIS 2 · SUPPORT

The normal force is ____ N because ____

EXHIBIT B · DRAW THE CRATE'S FORCES (dot + labeled arrows)

WITNESS · TRUE OR FALSE?

"After the rope snapped, the rope's force kept the crate sliding for a second."

THE VERDICT

Every force on the crate came from ____. When contact ends, ____

K2 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 10 · K2 · LOOP CHAIN

I have... who has? Forces

10 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, and keep going.

★ START — **A**

WHO HAS…

Weight of a 3 kg book (g ≈ 10 N/kg)?

I have **tension** — **B**

WHO HAS…

Mass is measured in…?

I have **normal force** — **C**

WHO HAS…

Weight of a 5 kg box with g = 9.8?

I have **friction** — **D**

WHO HAS…

A force that acts without touching?

I have **30 N** — **E**

WHO HAS…

The force a table pushes up with?

I have **gravity** — **F**

WHO HAS…

Is "the force of motion" a real force?

I have **49 N** — **G**

WHO HAS…

A rope's pull is called…?

I have **kilograms** — **H**

WHO HAS…

The force that opposes sliding?

I have **no — not an interaction** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

K2 1 2 3

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Contact, non-contact, or not a force?

10 N

DIRECTIONS · Write each card's letter in its column.

A the table's normal force	**B** Earth's gravity on a book	**C** the rope's tension	**D** velocity of a cart
E friction from the floor	**F** your hand's applied push	**G** mass of 3 kg	**H** gravity on 2026-XJ from the Sun
I air resistance	**J** "the force of motion"	**K** a magnet pulling a paperclip	**L** acceleration

CONTACT FORCE	NON-CONTACT FORCE	NOT A FORCE
needs touching	acts at a distance	a quantity, not a push/pull

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

K2 1 2 3

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Comic: same mass, different weight

10 CS

YOUR READER
A younger student who thinks mass and weight are the same thing.
`Stick figures are fine. The physics gets checked.`

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **ORANGE** — Weight arrows that change size (and none in deep space)
- ☐ ◯ **PENCIL** — "mass = ____ kg" the same in every panel
- ☐ ◯ **PENCIL** — Weight written in newtons (N)

1 · ON EARTH *An astronaut holding a backpack.*

CAPTION _____

2 · ON THE MOON *Same astronaut, same backpack.*

CAPTION _____

3 · DEEP SPACE *Far from any planet or moon.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **mass · weight · gravity · newton**

K2 (1) (2) (3)

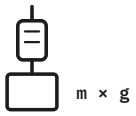
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Doodle Notes · weight

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Weight · peso (ES) · peso (PT)



TODAY'S CASE

The same backpack is carried from Earth to the Moon. The Moon's gravitational field is weaker.

01 · SKETCH AND LABEL

Draw the backpack twice. Give both drawings the same mass label in kilograms.

02 · CHANGE THE PICTURE

Add orange weight arrows with different lengths. Label their unit and name which measurement changed.

Annotate your sketches. Circle the mark that best shows weight. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The mass stays ___, but the weight …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else